



- 21 (c) Non-ideal solution
- 22 (a) Benzene–Chloroform
Explanation:
Benzene–chloroform shows negative deviation due to stronger intermolecular attraction.
- 23 (c) $H_2O + HCl$
Explanation:
Positive deviation occurs when A–B interactions are weaker than A–A and B–B interactions.
- 24 (d) All of these
Explanation:
For ideal solutions:
- $P_A = P_A^0 X_A$
 - $\Delta H_{mix} = 0$
 - $\Delta V_{mix} = 0$
- So all given statements are not true for ideal solutions.
- 25 (a) $\Delta S_{mix} = 0$
Explanation:
Entropy of mixing is not zero; it increases on mixing.
- 26 (b) Acetone–Benzene
Explanation:
Acetone–benzene shows positive deviation.
- 27 (a) An ideal solution
- 28 (c) Benzene + toluene
- 29 (a) Positive deviation from Raoult's law
Explanation:



Weaker intermolecular forces increase vapour pressure, producing positive deviation.

30 (a) The enthalpy of mixing is zero

Explanation:

For ideal solutions:

- $\Delta H_{mix} = 0$
- $\Delta V_{mix} = 0$
- $\Delta S_{mix} > 0$

